

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES				
10							***** VALVES *****			821	TO BE USED FOR LINES SUBJECT TO AIV OR WHERE REQUIRED BY STRESS.			
20		VB2					***** BALL VALVES *****			B1	ALL BOLTS ARE METRIC DIA.			
30										B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10			
30	U04	VA5	BALL	A01AC01	2	4	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT RB LO	0	0	B6	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AS22			
40	U04		BALL	A01AC03	6	6	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT RB LO	0	0	FR5	USE FLAT FACE FLANGES WHEN CONNECTING TO FLAT FACE PLASTIC OR FRP FLANGES.			
50	U04		BALL	A01AC07	8	8	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT RB GO	0	0	FS1	USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES.			
60	U04	VA5	BALL	BA6B2	0.5	0.5	BALL 150# SCRD FNPT BS1400 LG2 BODY AND TRIM FB LO	0	0	GF2	USE WITH FLAT FACE FLANGE ONLY.			
70			BALL	BA6B2	0.75	2	BALL 150# SCRD FNPT BS1400 LG2 BODY AND TRIM FB LO	0	0	SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.			
80	U04	VA5	BALL	B01AC01	3	3	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO	0	0	UD7	USE THREDOLET AND SCREWED PLUG FOR HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST			
90	U04	VA5	BALL	B01AC03	4	6	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO	0	0	UG3	PTFE GASKET ONLY TO BE USED FOR INTERFACE WITH POLYETHYLENE (HDPE) UNDERGROUND PIPING			
100	U04	VA5	BALL	B01AC07	6	8	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO	0	0	UG4	BOLTS ONLY TO BE USED FOR INTERFACE WITH POLYETHYLENE (HDPE) UNDERGROUND PIPING			
110	U04	VAW	BALL	B03AC01	0.75	0.75	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO	0	0	UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.			
120	U04		BALL	BB6B2	0.75	2	BALL 300# SCRD FNPT BS1400 LG2 BODY AND TRIM FB LO	0	0	VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS			
130							***** BUTTERFLY VALVES *****			VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.			
140			BUTTERFLY	T01AC61	10	24	BUTTERFLY 150# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO	0	0	VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.			
150			BUTTERFLY	T01JC61	26	36	BUTTERFLY 150# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO ASME B16.47 SERIES A	0	0	VC1	SPRING PISTON CHECK VALVES ARE PREFERRED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.			
160							***** GATE VALVES *****							
170	U04		GATE	GA6B1	0.5	2	GATE 150# SCRD FNPT BS1400 LG2 BODY AND TRIM HW	0	0					
180			GATE	G01AC41	3	8	GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW	0	0					
190	U04		GATE	G03AC41	3	3	GATE 300# RF LTCS A352 LCC TRIM 16 WEDGE HW	0	0					
200							***** GLOBE VALVES *****							
210			GLOBE	LA6B1	0.75	2	GLOBE 150# SCRD FNPT BS1400 LG2 BODY AND TRIM HW	0	0					
220			GLOBE	L01AC41	3	6	GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y HW	0	0					
230			GLOBE	L01AC42	8	10	GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y GO	0	0					
240							***** CHECK VALVES *****							
250														
250		VC1	CHECK	CA6B1	0.75	2	CHECK 150# SCRD FNPT BS1400 LG2 BODY AND TRIM SPRING PISTON	0	0					
260			CHECK	C01AC45	3	6	CHECK DUAL WAFER LUG 150# RF LTCS A352 LCC TRIM 16 THROU BOLTED	0	0					
270			CHECK	C01AC52	8	24	CHECK DUAL DBL FLG 150# RF LTCS A352 LCC TRIM 16	0	0					
280			CHECK	C01JC52	26	36	CHECK DUAL DBL FLG 150# RF LTCS A352 LCC TRIM 16 B16.47 SERIES A	0	0					
290							***** PIPE *****							
300														
300			PIPE	5358601	0.75	1.5	PIPE SCH 160 SMLS LTCS A333 GR.6 GALV A153 T&C	0	0					
310			PIPE	5358602	2	2	PIPE SCH XS SMLS LTCS A333 GR.6 GALV A153 T&C	0	0					
320			PIPE	5358602	4	4	PIPE SCH XS SMLS LTCS A333 GR.6 GALV A153 T&C	0	0					
330			PIPE	5358514	3	3	PIPE SCH XS SMLS LTCS A333 GR.6 BE	0	0					
340	U04		PIPE	5358514	3	3	PIPE SCH XS SMLS LTCS A333 GR.6 BE	0	0					
350			PIPE	5358505	4	14	PIPE SCH STD SMLS LTCS A333 GR.6 BE	0	0					
360			PIPE	5358501	16	16	PIPE SCH 20 SMLS LTCS A333 GR.6 BE	0	0					
370			PIPE	5358597	18	20	PIPE SCH 20 EFW LTCS A671 CC60 CL.22 BE	0	0					
380			PIPE	5358597	24	24	PIPE SCH 20 EFW LTCS A671 CC60 CL.22 BE	0	0					
390			PIPE	5358576	26	36	PIPE SCH STD EFW LTCS A671 CC60 CL.22 BE	0	0					
400							***** FITTINGS *****							
410														
410	U04		ELBOW	5491326	0.5	0.5	ELL 90 DEG 3000# SCRD MNPT X FNPT LTCS A350 LF2 CL.1 GALV A153	0	0					
420	U04		ELBOW	5491291	0.5	0.5	ELL 90 DEG 3000# SCRD MNPT LTCS A350 LF2 CL.1 GALV A153	0	0					
430	U04		ELBOW	5491290	0.5	2	ELL 90 DEG 3000# SCRD FNPT LTCS A350 LF2 CL.1 GALV A153	0	0					
440			ELBOW	5392600	3	3	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6	0	0					
450			ELBOW	5392438	4	14	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6	0	0					
460			ELBOW	5392276	16	16	ELL BW 90 DEG LR SCH 20 LTCS A420 WPL6	0	0					
470	U04	821	ELBOW	5392384	16	16	ELL 90 DEG LR SCH 40 LTCS A420 WPL6	0	0					
480			ELBOW	5392280	18	20	ELL BW 90 DEG LR SCH 20 LTCS A420 WPL6-W	0	0					
490			ELBOW	5392280	24	24	ELL BW 90 DEG LR SCH 20 LTCS A420 WPL6-W	0	0					
500			ELBOW	5392496	26	36	ELL 90 DEG LR SCH STD LTCS A420 WPL6-W	0	0					
510	U04		ELBOW	5491327	0.5	0.5	ELL 45 DEG 3000# SCRD MNPT X FNPT LTCS A350 LF2 CL.1 GALV A153	0	0					
520	U04		ELBOW	5491301	0.5	0.5	ELL 45 DEG 3000# SCRD MNPT LTCS A350 LF2 CL.1 GALV A153	0	0					
530	U04		ELBOW	5491300	0.5	2	ELL 45 DEG 3000# SCRD FNPT LTCS A350 LF2 CL.1 GALV A153	0	0					
540			ELBOW	5392606	3	3	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6	0	0					
550			ELBOW	5392444	4	14	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6	0	0					
560			ELBOW	5392282	16	16	ELL BW 45 DEG SCH 20 LTCS A420 WPL6	0	0					
570			ELBOW	5392281	18	20	ELL BW 45 DEG LR SCH 20 LTCS A420 WPL6-W	0	0					
580			ELBOW	5392281	24	24	ELL BW 45 DEG LR SCH 20 LTCS A420 WPL6-W	0	0					
590			ELBOW	5392607	26	36	ELL BW 45 DEG SCH STD LTCS A420 WPL6-W	0	0					
			CONSTRUCTION		CORR.ALLOW.		SERVICE	MATERIAL		DATE	BY	REV	SHEET	PIPING LINE CLASS
			2" & SMALLER - GALVANIZED, THREADED COUPLED 3" & ABOVE - BUTT WELD - GALVANIZED AFTER FABRICATION PWHT FOR WT > 19MM SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE		CORR. ALLOW. 3MM		SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: 15.8 BARG CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS + GALV FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 150 °C		12/16/2024	BY OZAC	U04	Page 1 of 4	150H2G

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES							
600																	
600	U04		TEE	5491310	0.75	2	TEE 3000# SCRD FNPT LTCS A350 LF2 CL.1 GALV A153	0	0								
610			TEE	5392609	3	3	TEE BW SCH XS LTCS A420 WPL6	0	0								
620			TEE	5392447	4	14	TEE BW SCH STD LTCS A420 WPL6	0	0								
630			TEE	5392285	16	16	TEE BW SCH 20 LTCS A420 WPL6	0	0								
640			TEE	5392295	18	20	TEE BW SCH 20 LTCS A420 WPL6-W	0	0								
650			TEE	5392295	24	24	TEE BW SCH 20 LTCS A420 WPL6-W	0	0								
660			TEE	5392405	26	36	TEE SCH STD LTCS A420 WPL6-W	0	0								
670	U04		TEE RED	5491311	0.5	1.5	TEE RED 3000# SCRD FNPT LTCS A350 LF2 CL.1 GALV A153	0.75	2								
680			TEE RED	5392454	3	3	TEE RED BW SCH STD/XS LTCS A420 WPL6	4	6								
690			TEE RED	5392445	4	12	TEE RED BW SCH STD LTCS A420 WPL6	6	14								
700			TEE RED	5392289	6	14	TEE RED BW SCH 20/STD LTCS A420 WPL6	16	16								
710			TEE RED	5392296	8	14	TEE RED BW SCH 20/STD LTCS A420 WPL6-W	18	24								
720			TEE RED	5392292	16	20	TEE RED BW SCH 20 LTCS A420 WPL6-W	18	24								
730			TEE RED	5556912	16	16	TEE RED BW SCH STD/ SCH 20 LTCS A420 WPL6-W	28	36								
740			TEE RED	5392428	28	28	TEE RED BW SCH STD LTCS A420 WPL6-W	36	36								
750																	
750	U04		CAP	5491330	0.75	2	CAP 3000# SCRD FNPT LTCS A350 LF2 CL.1 GALV A153	0	0								
760	U04	69	CAP	5491330	4	4	CAP 3000# SCRD FNPT LTCS A350 LF2 CL.1 GALV A153	0	0								
770			CAP	5392610	3	3	CAP BW SCH XS LTCS A420 WPL6	0	0								
780			CAP	5392448	4	14	CAP BW SCH STD LTCS A420 WPL6	0	0								
790			CAP	5392448	26	36	CAP BW SCH STD LTCS A420 WPL6	0	0								
800			CAP	5392286	16	20	CAP BW SCH 20 LTCS A420 WPL6	0	0								
810			CAP	5392286	24	24	CAP BW SCH 20 LTCS A420 WPL6	0	0								
820																	
820	U04		RED CONC	5392452	2.5	3	REDUCER CONC SCH STD/XS LTCS A420 WPL6	4	6								
830			RED CONC	5392449	4	12	REDUCER CONC SCH STD LTCS A420 WPL6	6	14								
840			RED CONC	5392290	8	14	REDUCER CONC SCH 20/STD LTCS A420 WPL6	16	16								
850			RED CONC	5392297	10	14	REDUCER CONC SCH 20/STD LTCS A420 WPL6-W	18	20								
860			RED CONC	5392287	16	20	REDUCER CONC SCH 20 LTCS A420 WPL6-W	18	24								
870			RED CONC	5392500	20	24	REDUCER CONC SCH STD/20 LTCS A420 WPL6-W	26	36								
880			RED CONC	5392505	26	32	REDUCER CONC SCH STD LTCS A420 WPL6-W	30	36								
890			RED ECC	5392453	3	3	REDUCER ECC SCH STD/XS LTCS A420 WPL6	4	6								
900			RED ECC	5392450	4	12	REDUCER ECC SCH STD LTCS A420 WPL6	6	14								
910			RED ECC	5392291	8	14	REDUCER ECC SCH 20/STD LTCS A420 WPL6	16	16								
920			RED ECC	5392298	10	14	REDUCER ECC SCH 20/STD LTCS A420 WPL6-W	18	20								
930			RED ECC	5392288	16	20	REDUCER ECC SCH 20 LTCS A420 WPL6-W	18	24								
940			RED ECC	5392508	20	24	REDUCER ECC SCH STD/20 LTCS A420 WPL6-W	26	36								
950			RED ECC	5392509	26	32	REDUCER ECC SCH STD LTCS A420 WPL6-W	30	36								
960																	
970							*****OLETS*****										
970	U04		OLETS	5504002	0.75	2	NIPOFLANGE 150# RF SCH STD/ XS LTCS A350 LF2 CL.1	4	14								
980	U04		OLETS	5504002	0.75	2	NIPOFLANGE 150# RF SCH STD/ XS LTCS A350 LF2 CL.1	26	36								
990	U04		OLETS	5504016	0.75	2	NIPOFLANGE 150# RF SCH 20/XS LTCS A350 LF2 CL.1	16	24								
1000	U04	UJ2	OLETS	5504025	1.5	1.5	NIPOFLANGE 300# RF SCH STD/160 LTCS A350 LF2 CL.1	4	14								
1010	U04	UJ2	OLETS	5504025	1.5	1.5	NIPOFLANGE 300# RF SCH STD/160 LTCS A350 LF2 CL.1	26	36								
1020	U04	UJ2	OLETS	5504019	1.5	1.5	NIPOFLANGE 300# RF SCH 20/160 LTCS A350 LF2 CL.1	16	24								
1030	U04		OLETS	5504020	1.5	1.5	NIPOFLANGE 300# RF SCH XS LTCS A350 LF2 CL.1	3	3								
1040																	
1040	U04	UD7	OLETS	5514772	0.5	2	THREDOLET 3000# LTCS A350 LF2 CL.1	3	36								
1050	U04		OLETS	5514773	0.5	1	THREDOLET FLAT BOTTOM 3000# LTCS A350 LF2 CL.1	0	0								
1060	U04		OLETS	5514773	2	2	THREDOLET FLAT BOTTOM 3000# LTCS A350 LF2 CL.1	0	0								
1070																	
1070	U04		OLETS	5505179	3	3	WELDOLET SCH STD/XS LTCS A350 LF2 CL.1	4	14								
1080	U04		OLETS	5505179	3	3	WELDOLET SCH STD/XS LTCS A350 LF2 CL.1	26	36								
1090	U04		OLETS	5505572	3	3	WELDOLET SCH 20/XS LTCS A350 LF2 CL.1	16	24								
1100	U04		OLETS	5505164	4	4	WELDOLET SCH STD LTCS A350 LF2 CL.1	8	14								
1110	U04		OLETS	5505164	4	12	WELDOLET SCH STD LTCS A350 LF2 CL.1	26	36								
1120	U04		OLETS	5505571	4	8	WELDOLET SCH 20/STD LTCS A350 LF2 CL.1	16	24								
1130																	
1130	U04	UJ2	ELBOFLG	5516623	1.5	1.5	ELBOFLANGE 300# RF SCH XS/160 LTCS A350 LF2 CL.1	3	3								
1140	U04	UJ2	ELBOFLG	5516625	1.5	1.5	ELBOFLANGE 300# RF SCH STD/160 LTCS A350 LF2 CL.1	4	14								
1150	U04	UJ2	ELBOFLG	5516625	1.5	1.5	ELBOFLANGE 300# RF SCH STD/160 LTCS A350 LF2 CL.1	26	36								
1160	U04	UJ2	ELBOFLG	5516619	1.5	1.5	ELBOFLANGE 300# RF SCH 20/160 LTCS A350 LF2 CL.1	16	24								
1170																	
1170							***** COUPLINGS, UNIONS & PLUGS *****										
1180	U04																
1180	U04		COUPLING	5491250	0.75	2	COUPLING 3000# SCRD FNPT LTCS A350 LF2 CL.1 GALV A153	0	0								
1190	U04		COUPLING	5491260	0.5	1.5	COUPLING RED 3000# SCRD FNPT LTCS A350 LF2 CL.1 GALV A153	0.75	2								
1200																	
1200	U04		UNION	5491270	0.75	2	UNION 3000# SCRD FNPT LTCS A350 LF2 CL.1 GALV A153	0	0								
1210	U04		UNION	5491271	0.75	2	UNION 3000# SCRD (MNPT/FNPT) LTCS A350 LF2 CL.1 GALV A153	0	0								
		TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION		CORR.ALLOW.		SERVICE		MATERIAL		DATE	BY	REV	SHEET	PIPING LINE CLASS
				2" & SMALLER - GALVANIZED, THREADED COUPLED 3" & ABOVE - BUTT WELD - GALVANIZED AFTER FABRICATION PWHT FOR WT > 19MM SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE		CORR. ALLOW. 3MM		SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: 15.8 BARG CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO		MATERIAL: LTCS + GALV FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 150 °C		12/16/2024	BY OZAC	U04	Page 2 of 4	150H2G	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES											
1220																					
1220	U04	UD7	FITTING SCRD	5491280	0.5	2	PLUG ROUND HEAD SCRD MNPT LTCS A350 LF2 CL.1 GALV A153	0	0												
1230																					
1230	U04		FITTING SCRD	5491325	0.5	1.5	BUSHING 3000# SCRD (MNPT/FNPT) HEX HEAD LTCS A350 LF2 CL.1 GALV A153	0.75	2												
1240																					
1240		FS1					***** FLANGES *****														
1250																					
1250	U04		TEE RED	5599691	0.5	1.5	FLG SCRD (FNPT) REDUCING 150# RF LTCS A350 LF2 CL.1 GALV A153	0.75	2												
1260	U04		FLANGE	5599691	2	2	FLG SCRD (FNPT) REDUCING 150# RF LTCS A350 LF2 CL.1 GALV A153	6	6												
1270	U04		TEE RED	5599702	0.5	1.5	FLG SCRD (FNPT) REDUCING 300# RF LTCS A350 LF2 CL.1 GALV A153	1	2												
1280	U04		FLANGE	5599689	0.5	2	FLG SCRD (FNPT) 150# RF LTCS A350 LF2 CL.1 GALV A153	0	0												
1290	U04	FR5	FLANGE	5599693	0.75	2	FLG SCRD (FNPT) 150# FF LTCS A350 LF2 CL.1 GALV A153	0	0												
1300	U04		FLANGE	5556264	2.5	3	FLG WN 150# RF LTCS A350 LF2 CL.1 SCH XS	0	0												
1310	U04	FR5	FLANGE	5556217	3	3	FLG WN 150# FF LTCS A350 LF2 CL.1 SCH XS	0	0												
1320	U04		FLANGE	5556261	4	14	FLG WN 150# RF LTCS A350 LF2 CL.1 SCH STD	0	0												
1330	U04	FR5	FLANGE	5556216	4	14	FLG WN 150# FF LTCS A350 LF2 CL.1 SCH STD	0	0												
1340	U04		FLANGE	5556265	16	20	FLG WN 150# RF LTCS A350 LF2 CL.1 SCH 20	0	0												
1350	U04		FLANGE	5556265	24	24	FLG WN 150# RF LTCS A350 LF2 CL.1 SCH 20	0	0												
1360	U04	FR5	FLANGE	5556220	16	16	FLG WN 150# FF LTCS A350 LF2 CL.1 SCH 20	0	0												
1370	U04		FLANGE	5556262	26	36	FLG WN 150# RF LTCS A350 LF2 CL.1 SCH STD B16.47 SERIES A	0	0												
1380	U04	FS1	FLANGE	5599701	0.75	2	FLG SCRD (FNPT) 300# RF LTCS A350 LF2 CL.1 GALV A153	0	0												
1390	U04	FS1	FLANGE	5599694	0.75	2	FLG SCRD (FNPT) 300# FF LTCS A350 LF2 CL.1 GALV A153	0	0												
1400	U04	FS1	FLANGE	5556476	3	3	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH XS	0	0												
1410	U04	FS1	FLANGE	5556455	3	3	FLG WN 300# FF LTCS A350 LF2 CL.1 SCH XS	0	0												
1420	U04	FS1	FLANGE	5556473	4	14	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH STD	0	0												
1430	U04	FS1	FLANGE	5556456	4	14	FLG WN 300# FF LTCS A350 LF2 CL.1 SCH STD	0	0												
1440	U04	FS1	FLANGE	5556474	16	20	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH 20	0	0												
1450	U04	FS1	FLANGE	5556474	24	24	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH 20	0	0												
1460	U04	FS1	FLANGE	5556457	16	20	FLG WN 300# FF LTCS A350 LF2 CL.1 SCH 20	0	0												
1470	U04	FS1	FLANGE	5556475	36	36	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH STD B16.47 SERIES A	0	0												
1480	U04	FS1	FLANGE	5556458	1.25	2	FLG SCRD (FNPT) PN16 RF LTCS A350 LF2 CL.1 GALV A153	0	0												
1490	U04		BLIND	5615567	0.5	24	FLG BLIND 150# RF LTCS A350 LF2 CL.1 GALV A153	0	0												
1500	U04		BLIND	5615575	26	36	FLG BLIND 150# RF LTCS A350 LF2 CL.1 GALV A153 B16.47 SERIES A	0	0												
1510	U04	FS1	BLIND	5615487	0.75	24	FLG BLIND 150# FF LTCS A350 LF2 CL.1 GALV A153	0	0												
1520	U04	FS1	BLIND	5615577	0.5	24	FLG BLIND 300# RF LTCS A350 LF2 CL.1 GALV A153	0	0												
1530	U04	FS1	FLANGE	5557033	6	6	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH STD	0	0												
1540	U04		FLANGE	5556270	4	6	FLG WN REDUCING 150# RF LTCS A350 LF2 CL.1 BORED TO SUIT SCH STD WT	6	8												
1550																					
1550							***** ORIFICE FLANGES *****														
1560																					
1560	U04		ORIFICE FLG	5623948	2	2	ORIF SET 300# RF SCRD (FNPT) LTCS A350 LF2 CL.1 GALV A153	0	0												
1570																					
1570	U04		ORIFICE FLG	5623947	3	3	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH XS	0	0												
1580	U04		ORIFICE FLG	5623944	4	14	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH STD	0	0												
1590	U04		ORIFICE FLG	5623941	16	24	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH 20	0	0												
1600	U04		ORIFICE FLG	5624801	36	36	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH STD ASME B16.47 SERIES A	0	0												
1610																					
1610							***** PIPE NIPPLES *****														
1620																					
1620			NIPPLE	5541200	0.5	1.5	NIPPLE SCH 160 LTCS A333 GR.6 TBE 3" LG (MNPT) GALV A153	0	0												
1630			NIPPLE	5541201	0.5	1.5	NIPPLE SCH 160 LTCS A333 GR.6 TBE 7" LG (MNPT) GALV A153	0	0												
1640			NIPPLE	5541214	2	2	NIPPLE SCH XS LTCS A333 GR.6 TBE 3" LG (MNPT) GALV A153	0	0												
1650			NIPPLE	5541204	2	2	NIPPLE SCH XS LTCS A333 GR.6 TBE 7" LG (MNPT) GALV A153	0	0												
1660		69	NIPPLE	5541193	4	4	NIPPLE SCH XS/160 LTCS A333-6 BOE/TOE 7" LG	0	0												
1670																					
1670							***** SWAGE NIPPLES *****														
1680																					
1680			RED CONC	5544150	0.75	1	SWAGE CONC SCH 160 LTCS A420 WPL6 TBE (MNPT) GALV A153	1	1.5												
1690			RED CONC	5544151	0.75	1.5	SWAGE CONC SCH XS/160 LTCS A420 WPL6 TBE (MNPT) GALV A153	2	2												
1700			RED CONC	5544142	0.75	1.5	SWAGE CONC SCH XS/160 LTCS A420 WPL6 BLE/TSE	3	3												
1710			RED CONC	5544154	0.75	1.5	SWAGE CONC SCH STD/160 LTCS A420 WPL6 BLE/TSE	4	4												
1720			RED CONC	5544092	2	2	SWAGE CONC SCH XS LTCS A420 WPL6 BLE/TSE	3	3												
1730			RED CONC	5544155	2	2	SWAGE CONC SCH STD/XS LTCS A420 WPL6 BLE/TSE	4	4												
1740			RED ECC	5549150	0.75	1	SWAGE ECC SCH 160 LTCS A420 WPL6 TBE (MNPT) GALV A153	1	1.5												
1750			RED ECC	5549151	0.75	1.5	SWAGE ECC SCH XS/160 LTCS A420 WPL6 TBE (MNPT) GALV A153	2	2												
1760			RED ECC	5549225	0.75	1.5	SWAGE ECC SCH XS/160 LTCS A420 WPL6 BLE/TSE	3	3												
1770			RED ECC	5549154	0.75	1.5	SWAGE ECC SCH STD/160 LTCS A420 WPL6 BLE/TSE	4	4												
1780			RED ECC	5549182	2	2	SWAGE ECC SCH XS LTCS A420 WPL6 BLE/TSE	3	3												
1790			RED ECC	5549155	2	2	SWAGE ECC SCH STD/XS LTCS A420 WPL6 BLE/TSE	4	4												
1800																					
1800		FS1					***** GASKETS *****														
1810																					
1810			GASKET	5673043	0.5	24	GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR	0	0												
	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN		CONSTRUCTION		CORR.ALLOW.		SERVICE	MATERIAL								DATE	BY	REV	SHEET	PIPING LINE CLASS	
			2" & SMALLER - GALVANIZED, THREADED COUPLED 3" & ABOVE - BUTT WELD - GALVANIZED AFTER FABRICATION PWHT FOR WT > 19MM SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE		CORR. ALLOW. 3 MM		SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: 15.8 BARG CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS + GALV FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 150 °C								12/16/2024	BY OZAC	U04	Page 3 of 4	150H2G	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1820	U04	FS1	GASKET	5673103	6	6	GASKET 600# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1830			GASKET	5674011	26	36	GASKET 150# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A	0	0	
1840		FS1	GASKET	5673033	0.5	24	GASKET 300# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1850		GF2	GASKET	5664562	0.75	16	GASKET 150# FF RING EPDM 3MM THK	0	0	
1860	U04	UG3	GASKET	5664201	4	12	GASKET 150# RF FLAT RING PTFE	0	0	
1870		FS1	GASKET	5674032	36	36	GASKET 300# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A	0	0	
1880		GF2	GASKET	5664563	0.5	24	GASKET 300# FF RING EPDM 3MM THICK	0	0	
1890										
1890		B1					***** BOLTING *****			
1900										
1900	U04		BOLTING	5675770	0.5	99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
1910	U04	UG4	BOLTING	5675768	0.5	99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) W/ 2 NUTS + LTCS WASHERS	0	0	
1920										
1920		SB6	BOLTING	5675675	12	24	JACK SCREW HEX HEAD A320 GR L7M	0	0	
1930										
1930		FS1					**** BLINDS & SPACERS ****			
1940										
1940			SPEC BLIND	5749582	0.75	14	SPECTACLE BLIND 150# RF LTCS A 516 GR.70 GALV A153	0	0	
1950		FS1	SPEC BLIND	5749583	0.75	10	SPECTACLE BLIND 300# RF LTCS A516 GR.70 GALV A153	0	0	
1960										
1960			PADDLE	5749942	16	24	PADDLE BLANK 150# RF LTCS A516 GR.70 GALV A153	0	0	
1970			PADDLE	5750025	28	36	PADDLE BLANK 150# RF LTCS A516 GR70 B16.47 SERIES A GALV A153	0	0	
1980			PADDLE	5750195	16	24	PADDLE SPACER 150# RF LTCS A516 GR.70 GALV A153	0	0	
1990			PADDLE	5750194	28	36	PADDLE SPACER 150# RF LTCS A516 GR.70 B16.47 SERIES A GALV A153	0	0	
2000		FS1	PADDLE	5749853	12	24	PADDLE BLANK 300# RF LTCS A516 GR.70 GALV A153	0	0	
2010		FS1	PADDLE	5750201	12	24	PADDLE SPACER 300# RF LTCS A516-GR70 GALV A153	0	0	
2020										

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		2" & SMALLER - GALVANIZED, THREADED COUPLED 3" & ABOVE - BUTT WELD - GALVANIZED AFTER FABRICATION PWHT FOR WT > 19MM SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: 15.8 BARG CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS + GALV FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 150 °C	12/16/2024	BY OZAC	U04	Page 4 of 4	150H2G